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Practice : 1

```
In [ ]: from transformers import BertTokenizer

reviews = [
    "This product performs efficiently and meets expectations",
    "The quality of the item is unsatisfactory",
    "Delivery was completed on time and packaging was secure",
    "The product stopped functioning after a short period",
    "Good value for the price",
    "Not worth the cost considering performance",
    "The product is reliable and functions as described",
    "The build quality is poor and feels fragile",
    "Customer support was responsive and helpful",
    "Battery performance is below expectations",
    "Overall experience with this product is satisfactory",
    "The design and appearance are well executed",
    "The product does not perform as specified",
    "Satisfied with the purchase",
    "Would not recommend this product",
    "Performance is consistent and easy to use",
    "The item arrived damaged and delayed",
    "Functions as expected without major issues",
    "Unsatisfactory experience with the product",
    "Durable and reliable under regular usage"
]

selected = [reviews[0], reviews[3], reviews[10]]

selected_lower = [r.lower() for r in selected]

tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')

encoded = tokenizer(selected_lower, padding=True, truncation=True, return_tensors='pt')

for i, review in enumerate(selected_lower):
    tokens = tokenizer.convert_ids_to_tokens(encoded['input_ids'][i])
    input_ids = encoded['input_ids'][i].tolist()
    attention_mask = encoded['attention_mask'][i].tolist()

    print(f"\nReview {i+1}: {review}")
    print(f"Tokens: {tokens}")
    print(f"Input IDs: {input_ids}")
    print(f"Attention Mask: {attention_mask}")
```

```
c:\Users\HP\AppData\Local\Programs\Python\Python312\Lib\site-packages\huggingface_hub\file_download.py:1142: FutureWarning: `resume_download` is deprecated and will be removed in version 1.0.0. Downloads always resume when possible. If you want to force a new download, use `force_download=True`.
  warnings.warn(
```

Review 1: this product performs efficiently and meets expectations
Tokens: ['[CLS]', 'this', 'product', 'performs', 'efficiently', 'and', 'meets', 'expectations', '[SEP]', '[PAD]']
Input IDs: [101, 2023, 4031, 10438, 18228, 1998, 6010, 10908, 102, 0]
Attention Mask: [1, 1, 1, 1, 1, 1, 1, 1, 1, 0]

Review 2: the product stopped functioning after a short period
Tokens: ['[CLS]', 'the', 'product', 'stopped', 'functioning', 'after', 'a', 'short', 'period', '[SEP]']
Input IDs: [101, 1996, 4031, 3030, 12285, 2044, 1037, 2460, 2558, 102]
Attention Mask: [1, 1, 1, 1, 1, 1, 1, 1, 1, 1]

Review 3: overall experience with this product is satisfactory
Tokens: ['[CLS]', 'overall', 'experience', 'with', 'this', 'product', 'is', 'satisfactory', '[SEP]', '[PAD]']
Input IDs: [101, 3452, 3325, 2007, 2023, 4031, 2003, 23045, 102, 0]
Attention Mask: [1, 1, 1, 1, 1, 1, 1, 1, 1, 0]

```
In [ ]: short_review = reviews[4].lower()
        long_review = reviews[2].lower()

        for label, review in [("Short", short_review), ("Long", long_review)]:
            tokens = tokenizer.tokenize(review)
            print(f"{label} Review: '{review}'")
            print(f"Token Length (including [CLS] and [SEP]): {len(tokens) + 2}\n")
```

Short Review: 'good value for the price'
Token Length (including [CLS] and [SEP]): 7

Long Review: 'delivery was completed on time and packaging was secure'
Token Length (including [CLS] and [SEP]): 11